

PROMPT ENGINEERING CHALLENGE

Introduction

Prompt engineering is the process of writing clear instructions for AI systems to get better results. In this assignment, the CREATE framework is used:

C – Context
R – Role
E – Example
A – Action
T – Tone
E – Expected Format

1. Summarising Notes

Prompt

Context: I am a college student preparing for exams.

Role: Act as an experienced teacher.

Example: Convert long notes into short points.

Action: Summarise the following notes on Artificial Intelligence into key points.

Tone: Simple and easy to understand.

Expected Format: Bullet points.

Notes:

Artificial Intelligence is a branch of computer science that enables machines to perform tasks requiring human intelligence. AI can learn from data, solve problems, understand language, and make decisions.

AI Response

- Artificial Intelligence (AI) is a branch of computer science.
- AI helps machines perform human-like tasks.
- AI can learn from data.
- AI can solve problems.
- AI can understand language.
- AI can make decisions.

- AI is used in many modern applications.

Why This Prompt Worked

The prompt clearly explained the role, action, and expected format. The AI understood that the information should be simplified into bullet points.

2. Generating a Study Plan

Prompt

Context: I am a second-year engineering student preparing for semester exams.

Role: Act as an academic mentor.

Example: Create a weekly study schedule.

Action: Generate a 7-day study plan for Data Structures and Python.

Tone: Motivating and practical.

Expected Format: Table with Day, Subject, and Study Hours.

AI Response

Day	Subject	Study Hours
Monday	Structures Basics	2 Hours
Tuesday	Arrays and Linked Lists	2 Hours
Wednesday	Python Functions	2 Hours
Thursday	Python OOP Concepts	2 Hours
Friday	Stack and Queue Problems	2 Hours
Saturday	Practice Coding Questions	3 Hours
Sunday	Revision and Mock Test	3 Hours

Why This Prompt Worked

The prompt provided clear context and a specific output format. The AI generated a structured and realistic study plan.

3. Creating Interview Questions

Prompt

Context: I am preparing for a software engineering internship interview.

Role: Act as a technical interviewer.

Example: Ask beginner-friendly questions.

Action: Generate 10 interview questions on Python programming.

Tone: Professional.

Expected Format: Numbered list.

AI Response

1. What is Python?
2. What are Python data types?
3. What is a list in Python?
4. What is the difference between a list and a tuple?
5. What is a function?
6. What is object-oriented programming?
7. What is inheritance?
8. What is exception handling?
9. What are Python modules?
10. Why do you want to learn Python?

Why This Prompt Worked

The role and context guided the AI to create relevant interview questions. The expected format ensured a clean numbered list.

4. Writing a Resume Summary

Prompt

Context: I am a Computer Science student looking for internships.

Role: Act as a professional resume writer.

Example: Write a short and attractive resume summary.

Action: Create a resume summary for a student skilled in Python, C, Git, and AI basics.

Tone: Professional and confident.

Expected Format: One paragraph.

AI Response

Motivated Computer Science student with knowledge of Python, C programming, Git, and Artificial Intelligence fundamentals. Passionate about learning new technologies and solving real-world problems. Strong teamwork and communication skills with a commitment to continuous improvement and professional growth.

Why This Prompt Worked

The prompt clearly defined the user's background and skills. The AI produced a concise and professional summary suitable for a resume.

5. Creating a Lesson Plan

Prompt

Context: I am teaching beginners about Python loops.

Role: Act as an experienced programming instructor.

Example: Include objectives, activities, and assessment.

Action: Create a one-hour lesson plan on Python loops.

Tone: Educational and engaging.

Expected Format:

- Learning Objectives
- Activities
- Assessment
- Conclusion

AI Response

Learning Objectives:

- Understand the concept of loops.
- Learn for loops and while loops.
- Write simple loop programs.

Activities:

- Introduction to loops (10 minutes)
- Demonstration of for loops (15 minutes)
- Demonstration of while loops (15 minutes)
- Hands-on coding practice (15 minutes)

Assessment:

- Write a program to print numbers from 1 to 10.
- Create a multiplication table using loops.

Conclusion:

Students will understand how loops work and will be able to write simple Python programs using loops.

Why This Prompt Worked

The CREATE framework provided clear instructions. The AI generated a complete lesson plan with objectives, activities, assessment, and conclusion.